

UrbanFix – Local Skilled Worker Booking System

Project Introduction

Project Overview

UrbanFix is a web-based platform that connects skilled workers with customers in urban areas who need their services. It solves the problem of finding reliable and trustworthy skilled workers by providing a centralized digital marketplace with transparent pricing, real-time booking, and professional profile management.

Problem Addressed

- Difficulty finding reliable skilled workers in urban areas
 - Lack of transparency in pricing and availability
 - No centralized system for worker scheduling and booking
 - Customers have no way to verify worker quality or experience
 - Workers lack a digital platform to showcase their services
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User Types

1. **Customers** - Browse, search, and book skilled workers
 2. **Workers** - Register their services, manage bookings, and maintain profiles
 3. **Administrators** - Manage the platform, oversee users, and maintain quality
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Service Categories Offered

- Electrician
 - Carpenter
 - AC Repairing
 - Painting
 - Mechanics
 - Plumber
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Key Features Implemented

Authentication & User Management

- Secure registration and login with email verification
- Separate login systems for customers and workers
- Password encryption using bcrypt
- JWT token-based authentication

Worker Search & Discovery

- Category-wise worker filtering
- Budget-based worker selection (search within price range)
- Browse all workers on the platform
- View detailed worker profiles

Booking System

- Real-time booking confirmation
- Track active and historical bookings
- Request management system for workers
- Instant service bookings

Dashboards

- Customer Dashboard - View bookings and worker details
- Worker Dashboard - Manage availability and view incoming bookings
- Admin Dashboard - Platform oversight and management

Communication

- Contact page with form validation
- Email notifications using Nodemailer
- FAQ accordion section
- Social media links

User Profiles

- Customer profile management and booking history
- Worker profile creation and skill management
- Update worker availability

Technology Stack

Frontend:

- React.js with Vite
- Tailwind CSS for styling
- Bootstrap for responsive components
- Framer Motion for animations
- React Router for navigation

Backend:

- Node.js runtime
- Express.js framework
- REST API architecture
- Nodemailer for email services

Database:

- MongoDB for data storage
- Mongoose for object modeling

Security & Validation:

- JWT for authentication
- Bcrypt/Bcryptjs for password hashing
- Joi for input validation
- CORS enabled for security

Development Tools:

- Nodemon for development server
- npm for package management
- Postman for API testing

API Routes

Route	Purpose
/auth	User authentication and registration
/workers	Worker profiles and management
/admin	Administrative functions
/contact	Contact form submissions

System Architecture

UrbanFix follows a **three-tier architecture**:

1. **Presentation Tier (Frontend)** - React.js-based responsive UI
 2. **Application Tier (Backend)** - Express.js REST API server
 3. **Data Tier** - MongoDB database
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Current Project Status

Completed Features

- Core booking system
- User authentication and authorization
- Worker search and filtering
- Real-time booking management
- Contact page with email notifications
- Admin dashboard setup
- Customer and Worker dashboards
- Category-wise worker browsing



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Project Information

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Project Title: UrbanFix – Local Skilled Worker Booking System

Key Technologies Summary

Layer	Technologies
Frontend	React, Vite, Tailwind CSS, Bootstrap, Framer Motion
Backend	Node.js, Express.js
Database	MongoDB, Mongoose
Security	JWT, Bcrypt, Joi Validation
Communication	Nodemailer

This comprehensive platform provides a complete solution for connecting skilled workers with customers in need of their services.